DATA TYPES, VARIABLES, AND OPERATORS IN PYTHON

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LEARNING OUTCOMES

- Define variables and explain their purpose
- Identify and use different data types in Python
- Apply operators to manipulate data
- Write simple programs using variables, data types, and operators

WHAT IS A VARIABLE?

- A variable is a named storage location that holds a value
- In Python, the type is decided automatically
- Example: age = 20, price = 99.50, grade = 'A', name = "Mary"

RULES FOR NAMING VARIABLES

- Must start with a letter or underscore (_)
- Can contain letters, digits, underscores
- No spaces or special characters
- Cannot use Python keywords (e.g., for, while, class)

DATA TYPES IN PYTHON

- int → Whole numbers (10, -25)
- float → Decimal numbers (12.5, -3.14)
- str → Text/characters ('Hello', 'A')
- bool → True/False values

EXAMPLE OF DATA TYPES

- age = 19 # integer
- height = 5.8 # float
- grade = 'B' # string
- is_student = True # boolean
- print(age, height, grade, is_student)

OPERATORS IN PYTHON

- Arithmetic Operators → + - * / % // **
- Relational Operators → < > == != <= >=
- Logical Operators → and, or, not
- Assignment Operators → = += -= *= /=

ARITHMETIC OPERATORS EXAMPLE

- a = 10, b = 3
- a + b = 13
- a - b = 7
- a * b = 30
- a / b = 3.33
- a // b = 3 (integer division)
- a % b = 1 (remainder)
- a ** b = 1000 (power)

RELATIONAL & LOGICAL OPERATORS

- $a = 5, b = 10$
- $a > b \rightarrow$ False
- $a < b \rightarrow$ True
- $x =$ True, $y =$ False
- x and y $\rightarrow$ False
- x or y $\rightarrow$ True
- not x $\rightarrow$ False

ASSIGNMENT OPERATORS

- a = 5
- a += 2 → 7
- a *= 3 → 21
- print(a)

CLASS ACTIVITY

- Create variables for your name, age, and average marks
- Print them using print()

PRACTICAL TASK 1

Write a program that asks the user to enter two numbers, then print:

- Their sum
- Their difference
- Their product
- Their quotient

PRACTICAL TASK 2

Write a program that takes two integers and prints the **remainder** when the first number is divided by the second.

PRACTICAL TASK 3

Ask the user to enter the **radius** of a circle. Compute and print the **area** of the circle:

PRACTICAL TASK 4

- Ask the user to input **principal**, **rate**, and **time**. Compute and print the **simple interest** using:

$$SI=P \times R \times T/100$$

PRACTICAL TASK 5

Write a program that swaps the values of two variables and prints them **before and after swapping**.

SUMMARY

- Variables store data in memory
- Data types define the kind of data stored
- Operators allow data manipulation
- They form the building blocks of Python programming